

|           |                                                                     |             |
|-----------|---------------------------------------------------------------------|-------------|
|           |                                                                     |             |
| 3(a)(i)   | <b>0.050 m</b>                                                      | <b>A1</b>   |
| 3(a)(ii)  | $\omega = v_o / x_o$                                                | <b>C1</b>   |
|           | $T = 2\pi / \omega$                                                 | <b>C1</b>   |
|           | $0.42 = (2\pi \times 0.050) / T$                                    |             |
|           | $T = \mathbf{0.75\ s}$                                              | <b>A1</b>   |
| 3(a)(iii) | one point labelled P where ellipse crosses displacement axis marked | <b>A1</b>   |
| 3(b)(i)   | (induced) e.m.f. proportional to rate                               | <b>M1</b>   |
|           | of change of (magnetic) flux (linkage)                              | <b>A1</b>   |
| 3(b)(ii)  | (there is) current in the circuit                                   | <b>B1</b>   |
|           | <i>either</i>                                                       |             |
|           | current causes thermal energy (dissipated) in resistor              | <b>B1</b>   |
|           | thermal energy comes from energy of magnet                          | <b>B1</b>   |
|           | <i>or</i>                                                           |             |
|           | current causes magnetic field around coil                           | <b>(B1)</b> |
|           | two fields cause an opposing force on magnet                        | <b>(B1)</b> |